CareBridge Project Report - Sample PDF

Chapter 1: Introduction

CareBridge is a healthcare management system written in modern C++. Patients can register, find nearby hospitals, browse doctors and book appointments. Doctors can triage appointment requests and maintain electronic health records for their patients.

The system serves two hospitals in the demo data set: CityCare General Hospital and Riverbend Medical Center.

Chapter 2: Data Structures Used

A singly linked list stores patients, doctors and appointments, giving constant-time insertion but linear-time search.
A priority queue orders appointment requests so that Emergency cases are seen first, then Urgent, then Routine requests.
A queue delivers patient notifications in first-in first-out order, while a stack tracks each doctor’s recently viewed patients.
Merge sort ranks hospitals by distance in O(n log n) time.

Chapter 3: Results and Future Work

The prototype handles one thousand synthetic patients with ease. Booking a slot and confirming it takes under fifty milliseconds. Future work includes a real database such as SQLite or Postgres, proper password hashing with bcrypt, and a mobile application. Telemedicine video calls are planned for the second release, scheduled for March 2027 according to the project roadmap.